



Properties

$$E(0) = B(0) = H(0) = S(0) = C(0) = 1$$

		T^+	T^-	k_D	k_H	k_σ	k_U	SI	Cost (USD/kg)
$K =$	MildSteelTemperate	400	0	50.0	high	1.20	0	0.30	0.80 – 1.20
	MildSteelColdWeather	400	−20	50.0	high	1.20	0	0.30	0.80 – 1.20
	AH36	400	0	48.0	mod	1.00	0	0.52	1.00 – 2.00
	DH36	400	−20	48.0	mod	1.00	0	0.52	1.00 – 2.00
	EH36	400	−40	48.0	mod	1.00	0	0.52	1.00 – 2.00
	EH40	400	−40	45.0	mod	0.95	0	0.58	1.50 – 2.50
	Al5083	300	−196	85.0	low	0.15	0	0.38	3.00 – 5.00
	Al5086	300	−196	85.0	low	0.15	0	0.38	3.00 – 5.00
	GRP	250	−100	40.0	mod	0.03	4.10	0.40	10.00 – 20.00
	CFRP	350	−150	20.0	mod	0.02	3.00	0.68	25.00 – 50.00
	TiGrade5	600	−196	55.0	neg	0.00	0	0.82	75.00 – 150.00
	RubberCompound	100	−40	120.0	high	0.00	2.50	0.03	2.00 – 5.00
	NitrileRubber	120	−30	110.0	low	0.00	2.00	0.03	3.00 – 6.00
	PolyurethaneCoating	150	−50	85.0	low	0.00	1.50	0.05	10.00 – 20.00
	Polyimide	350	−269	60.0	mod	0.00	3.00	0.08	80.00 – 150.00
	Polypropylene	130	−10	150.0	low	0.00	4.00	0.04	1.50 – 3.00
	KevlarComposite	300	−100	25.0	high	0.05	3.50	0.60	30.00 – 60.00
	CeramicComposite	1400	−150	6.0	neg	0.00	0	0.72	50.00 – 120.00
	TungstenCarbide	800	−150	5.5	neg	0.001	0	0.88	54.00 – 79.00
	SS304	870	−196	45.0	low	3.12	0	0.45	3.00 – 5.00
	SS316	870	−196	45.0	low	1.98	0	0.48	4.00 – 6.00
	ChromolySteel	530	−60	36.1	mod	0.85	0	0.65	2.00 – 4.00
	CastIron	600	−20	40.0	high	1.50	0	0.35	1.00 – 1.50
	NickelAlloy625	982	−196	40.0	neg	0.061	0	0.78	35.00 – 50.00
	Silicon	1414	−200	12.0	neg	0.00	0	0.12	15.00 – 25.00
	Germanium	938	−200	14.0	neg	0.00	0	0.10	1340.00
	GalliumArsenide	1238	−200	15.0	neg	0.00	0	0.10	2269.40
	SapphireGlass	2000	−200	8.0	neg	0.00	0	0.65	100.00 – 200.00
	BorosilicateGlass	500	−200	35.0	neg	0.00	0	0.20	15.00 – 30.00
	PhosphateEsterFluid	200	−20	—	vhigh	—	0	0.00	15.00 – 25.00/L
	LithiumCompound	180	−30	—	high	—	0	0.00	30.00 – 50.00
	PiezoelectricCeramic	350	−150	18.0	high	0.01	0	0.15	150.00 – 300.00

$$K = \begin{bmatrix} & T^+ & T^- & D & H & \sigma & U \\ E & k_{E,T^+} & k_{E,T^-} & k_{E,D} & 0 & 0 & 0 \\ B & k_{B,T^+} & k_{B,T^-} & 0 & 0 & 0 & k_{B,U} \\ H & k_{H,T^+} & k_{H,T^-} & 0 & 0 & 0 & k_{H,U} \\ S & 0 & 0 & k_{S,D} & k_{S,H} & 0 & k_{S,U} \\ C & 0 & 0 & 0 & k_{C,H} & k_{C,\sigma} & 0 \end{bmatrix}$$

Environmental Factors

1. Extreme heat

1. Overview
 1. Engine: oil thins, cooling system overloads, intake air is less dense reducing power
 2. Battery: permanent capacity reduction
 3. Hydraulics: fluid thins, seals degrade
2. Input
 1. $T(t)$ - temperature as function of time
3. Output
 1. $\frac{dE}{dt} = -k_E \cdot \max(T(t) - T_E, 0)$
 2. $\frac{dB}{dt} = -k_B \cdot \max(T(t) - T_B, 0)$
 3. $\frac{dH}{dt} = -k_H \cdot \max(T(t) - T_H, 0)$

2. Extreme cold

1. Overview
 1. Engine: cold starts fail, oil thickens, fuel gels
 2. Battery: immediate capacity drop, charge acceptance drops
 3. Hydraulics: fluid sluggish, actuators fail
2. Input
 1. $T(t)$ - temperature as function of time
3. Output
 1. $\frac{dE}{dt} = -k_E \cdot \max(T_E - T(t), 0)$
 2. $\frac{dB}{dt} = -k_B \cdot \max(T_B - T(t), 0)$
 3. $\frac{dH}{dt} = -k_H \cdot \max(T_H - T(t), 0)$

3. Dust ingestion

1. Overview
 1. Engine: intake filters clog, abrasive particles score cylinder walls
 2. Sensors: optical surfaces scratched and obscured, radar antenna elements clogged
2. Input
 1. $D(t)$ - particulate concentration as function of time
3. Output
 1. $\frac{dE}{dt} = -k_E \cdot D(t)$

$$2. \quad \frac{dS}{dt} = -k_S \cdot D(t)$$

4. Humidity

1. Overview

1. Sensors: condensation on optics, corrosion on electrical contacts
2. Chassis: rust at weld points and joints, seal degradation on hatches

2. Input

1. $Y(t)$ - relative humidity as function of time

3. Output

1. $\frac{dS}{dt} = -k_S \cdot \max(Y(t) - Y_S, 0)$
2. $\frac{dC}{dt} = -k_C \cdot \max(Y(t) - Y_C, 0)$

5. Salinity

1. Overview

1. Chassis: accelerated corrosion at welds, joints, underbelly

2. Input

1. $\sigma(t)$ - salinity concentration as function of time

3. Output

1. $\frac{dC}{dt} = -k_C \cdot \sigma(t)^2$

6. UV / Solar radiation

1. Overview

1. Sensors: optical coating degradation, accuracy drift
2. Hydraulics: external hoses become brittle
3. Battery: permanent capacity reduction from sustained heat load

2. Input

1. $U(t)$ - irradiance as function of time

3. Output

1. $\frac{dS}{dt} = -k_S \cdot U(t)$
2. $\frac{dH}{dt} = -k_H \cdot U(t)$
3. $\frac{dB}{dt} = -k_B \cdot U(t)^2$

Vehicle Health

$$V(t) = \frac{4}{13} \cdot E(t) + \frac{3}{13} \cdot B(t) + \frac{3}{13} \cdot H(t) + \frac{1}{13} \cdot S(t) + \frac{2}{13} \cdot C(t)$$